Methods and Databases in Empirical Finance - Sample Exam Questions

Based on Lectures 1-12 and Exercises 1-9

Instructions:

- These sample questions follow the Mock Exam format
- 1 point = approximately 1 minute to answer
- Maximum question size: 8 points (8 minutes)
- Mix of open questions, calculations, and multiple-choice questions (MCQ)
- At least 4 questions per major topic

Lecture 1 & Exercise 1: Introduction to Empirical Finance

Question 1.1 - Capital Markets Fundamentals (3 points)

Explain the fundamental capital allocation problem that capital markets solve. In your answer, describe why direct financing through capital markets is more efficient than relying solely on financial intermediaries for certain types of investments.

Question 1.2 - Market Types (3 points)

Distinguish between primary and secondary markets. Provide one example of a transaction in each market and explain why secondary markets are essential for the functioning of primary markets.

Question 1.3 - Risk-Return Relationship (4 points)

Consider two portfolios:

- Portfolio A: Expected return = 8%, Standard deviation = 12%
- Portfolio B: Expected return = 6%, Standard deviation = 8%
- Risk-free rate = 2%

1. Calculate the Sharpe ratio for both portfolios (2 points)
2. Which portfolio offers better risk-adjusted performance? (1 point)
3. What does the Sharpe ratio measure economically? (1 point)

Question 1.4 - MCQ: Market Efficiency (2 points)

According to the Efficient Market Hypothesis (EMH), which form of market efficiency implies that technical analysis cannot generate abnormal returns?

A) Strong-form efficiency
B) Semi-strong form efficiency
C) Weak-form efficiency
D) All forms of efficiency

Lecture 2 & Exercise 2: Financial Data Structures

Question 2.1 - CRSP Data Filters (6 points)

You are constructing a sample of US common stocks from CRSP. Explain the purpose of applying the following filters and what would happen if each filter were omitted:

1. Share Type = "NS" (common shares) (2 points)
2. US Incorporation Flag = "Y" (2 points)
3. Primary Exchange = "N", "A", "Q" (NYSE, AMEX, NASDAQ) (2 points)

Question 2.2 - Identifiers (4 points)

Explain the difference between PERMNO and PERMCO in CRSP data. Why might a single PERMCO be associated with multiple PERMNOs? Provide a practical example.

Question 2.3 - Look-Ahead Bias (4 points)

You want to form portfolios at the end of each month based on firm size (market capitalization).

1. Which month's price and shares outstanding should you use to calculate market cap for portfolio formation at the end of January 2024? (2 points)
2. Explain why using end-of-February 2024 prices would introduce look-ahead bias. (2 points)

Question 2.4 - MCQ: Return Calculations (2 points)

Why do researchers typically use returns instead of price levels in empirical finance?

A) Returns are easier to calculate than prices
B) Returns are scale-free and comparable across stocks
C) Prices contain more noise than returns
D) Returns are always normally distributed

Question 2.5 - Compustat Matching (5 points)

1. Explain why fiscal year-end dates in Compustat (datadate) do not always align with calendar year-end. (2 points)
2. Why is it important to use only the last available datadate for each fiscal year when merging CRSP and Compustat data? (3 points)

Lecture 3 & Exercise 3: Mean-Variance Portfolio Theory

Question 3.1 - Portfolio Variance Calculation (6 points)

Consider a two-asset portfolio with the following characteristics:

- Asset 1: Expected return = 10%, Standard deviation = 20%, Weight = 60%
- Asset 2: Expected return = 6%, Standard deviation = 10%, Weight = 40%
- Correlation between Asset 1 and Asset 2 = 0.3

1. Calculate the expected return of the portfolio. (2 points)
2. Calculate the variance of the portfolio. (3 points)
3. Calculate the standard deviation of the portfolio. (1 point)

Question 3.2 - Efficient Frontier Concepts (5 points)

1. Define the efficient frontier graphically and describe what it represents. (3 points)
2. Why would a rational investor never choose a portfolio below the efficient frontier? (2 points)

Question 3.3 - Minimum Variance Portfolio (4 points)

Given two assets with:

- $\sigma_1^2 = 0.04$, $\sigma_2^2 = 0.01$, $\sigma_{12} = 0.01$

Calculate the weight of Asset 1 in the minimum variance portfolio using the formula:

$w_1 = (\sigma_2^2 - \sigma_{12}) / (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12})$

Question 3.4 - MCQ: Capital Market Line (2 points)

What does the Capital Market Line (CML) represent?

A) The relationship between firm size and expected returns
B) The optimal trade-off between risk and return using the risk-free asset and the tangency portfolio

C) The minimum variance frontier for risky assets only
D) The relationship between beta and expected returns

Question 3.5 - Short Selling (3 points)

Explain how short selling works. What does it mean for a portfolio weight to be negative, and what constraint does this violate in basic mean-variance optimization?

Lecture 4 & Exercise 4: CAPM and Factor Models

Question 4.1 - CAPM Calculation (5 points)

You have the following information for Stock A:

- Covariance with market = 0.024
- Market variance = 0.04
- Risk-free rate = 3%
- Expected market return = 9%

1. Calculate the beta of Stock A. (2 points)
2. Calculate the expected return according to CAPM. (2 points)
3. If the realized return is 8%, what is the alpha? (1 point)

Question 4.2 - CAPM Assumptions (4 points)

List and briefly explain four key assumptions of the Capital Asset Pricing Model (CAPM). For each assumption, state whether it is realistic in practice.

Question 4.3 - Systematic vs. Idiosyncratic Risk (4 points)

1. Define systematic risk and idiosyncratic risk. (2 points)
2. According to CAPM, which type of risk is rewarded with a risk premium and why? (2 points)

Question 4.4 - Fama-French Three-Factor Model (5 points)

The Fama-French three-factor model is:

$E[R_i] - R_f = \alpha_i + \beta_i(E[R_m] - R_f) + s_i E[SMB] + h_i E[HML]$

1. Explain what SMB and HML represent. (3 points)
2. Why were these factors added beyond the market factor? (2 points)

Question 4.5 - MCQ: Risk Premium (2 points)

Which type of risk is rewarded with a risk premium according to the CAPM?

A) Idiosyncratic risk that is unique to a firm
B) Total risk measured by return variance
C) Systematic risk that cannot be diversified away
D) Liquidity risk arising from trading frictions

Lecture 5 & Exercise 5: Behavioral Finance

Question 5.1 - Behavioral Biases (6 points)

Name and explain three behavioral biases discussed in the lecture. For each bias:

1. Provide a clear definition (underline the bias name)
2. Give a practical example of how it affects investor decision-making

Question 5.2 - Overconfidence (4 points)

1. Define overconfidence bias. (2 points)
2. Explain how overconfidence can lead to excessive trading and poor portfolio performance. (2 points)

Question 5.3 - Prospect Theory (5 points)

1. Describe the key difference between expected utility theory and prospect theory. (2 points)
2. Explain the concept of loss aversion and how it affects investor behavior. (3 points)

Question 5.4 - MCQ: Disposition Effect (2 points)

The disposition effect refers to investors' tendency to:

A) Hold winning stocks too long and sell losing stocks too quickly
B) Sell winning stocks too quickly and hold losing stocks too long
C) Always follow the herd in investment decisions
D) Overestimate their ability to predict market movements

Question 5.5 - Limits to Arbitrage (4 points)

Explain two reasons why arbitrage opportunities may persist in financial markets even when they are identified by sophisticated investors.

Lecture 6 & Exercise 6: Performance Evaluation

Question 6.1 - Jensen's Alpha Calculation (6 points)

A mutual fund has the following characteristics:

- Average monthly return = 1.2%
- Beta = 1.3
- Average risk-free rate = 0.3%
- Average market excess return = 0.6%

1. Calculate Jensen's alpha. (3 points)
2. Interpret the result - did the fund outperform or underperform? (2 points)
3. Is this alpha statistically different from zero if the t-statistic is 1.5? (Use critical value of 1.96 for 5% significance) (1 point)

Question 6.2 - Information Ratio (4 points)

1. Define the Information Ratio and write its formula. (2 points)
2. Explain how it differs from the Sharpe ratio and when it is more appropriate to use. (2 points)

Question 6.3 - Treynor Ratio (3 points)

The Treynor ratio is calculated as $(R_p - R_f) / \beta_p$. Explain what this ratio measures and how it differs from the Sharpe ratio in terms of the risk measure used.

Question 6.4 - MCQ: Performance Attribution (2 points)

Which performance measure adjusts for systematic risk by comparing portfolio returns to a benchmark with the same beta?

A) Sharpe ratio
B) Jensen's alpha
C) Sortino ratio
D) Maximum drawdown

Question 6.5 - Tracking Error (4 points)

1. Define tracking error. (2 points)
2. A fund has an information ratio of 0.5 and an alpha of 2% per year. Calculate the tracking error. (2 points)

Lecture 7 & Exercise 7: Trading Strategies and Anomalies I

Question 7.1 - Portfolio Sorts (6 points)

Explain the four-step framework for constructing portfolios based on firm characteristics:

1. Step 1: Rank stocks
2. Step 2: Form portfolios
3. Step 3: Compute returns
4. Step 4: Test for abnormal performance

For each step, briefly describe what is done.

Question 7.2 - Long-Short Strategies (5 points)

1. Explain what a zero-investment long-short portfolio is. (2 points)
2. Why is this approach market-neutral in theory? (2 points)
3. If the long portfolio returns 15% and the short portfolio returns 5%, what is the strategy return? (1 point)

Question 7.3 - Univariate vs. Bivariate Sorts (5 points)

Explain the difference between independent and dependent bivariate portfolio sorts. When would you use each approach? Provide an example for each.

Question 7.4 - Fama-MacBeth Regression (6 points)

Explain the Fama-MacBeth regression methodology:

1. Describe what is done in Step 1 (cross-sectional regressions). (3 points)
2. Describe what is done in Step 2 (time-series averaging). (2 points)
3. Why is this two-step procedure necessary? (1 point)

Question 7.5 - MCQ: NYSE Breakpoints (2 points)

Why do researchers use NYSE breakpoints when forming portfolios, even when the sample includes NASDAQ stocks?

A) NYSE stocks are more liquid than NASDAQ stocks
B) NASDAQ stocks are typically smaller, and using all-stock breakpoints would create size-biased portfolios
C) NYSE has stricter listing requirements
D) NASDAQ data is less reliable

Lecture 8: Trading Strategies and Anomalies II

Question 8.1 - Factor Zoo Problem (5 points)

1. Explain what the "factor zoo" problem refers to in empirical asset pricing. (2 points)
2. Why has the number of documented factors grown so dramatically since the 1970s? Provide two reasons. (3 points)

Question 8.2 - P-Hacking (5 points)

1. Define p-hacking in the context of financial research. (2 points)
2. Give two specific examples of how researchers might engage in p-hacking when testing trading strategies. (3 points)

Question 8.3 - Multiple Testing (4 points)

Harvey, Liu, and Zhu (2016) suggest using a t-statistic hurdle of 3.0 instead of the traditional 1.96. Explain why this higher threshold is recommended in modern asset pricing research.

Question 8.4 - Post-Publication Decline (6 points)

McLean and Pontiff (2016) document that anomaly returns decline by 58% after publication.

1. What are the two main explanations for this decline? (4 points)
2. Which explanation is more consistent with market efficiency? (2 points)

Question 8.5 - MCQ: Replication (2 points)

According to Hou, Xue, and Zhang (2020), what percentage of published anomalies fail to replicate when using the higher multiple-testing hurdle of $t > 2.78$?

A) 35%
B) 50%
C) 65%
D) 82%

Lecture 9 & Exercise 8: Mutual Funds and Active Management

Question 9.1 - Mutual Funds vs. ETFs (6 points)

Compare mutual funds and Exchange-Traded Funds (ETFs) along three dimensions:

1. Trading and pricing mechanism (2 points)
2. Fee structure (2 points)
3. Tax efficiency (2 points)

Question 9.2 - Berk-Green Model (5 points)

Explain the Berk-Green (2004) equilibrium model of active management:

1. What happens when a skilled manager is identified? (2 points)
2. Why does alpha eventually go to zero? (2 points)
3. Why do investors still allocate capital to active managers in equilibrium? (1 point)

Question 9.3 - Luck vs. Skill (5 points)

Fama and French (2010) find that aggregate mutual fund alpha is close to zero before fees and negative after fees.

1. What does this imply about the average skill level of mutual fund managers? (2 points)
2. How can we distinguish luck from skill in fund performance? (3 points)

Question 9.4 - Active Share (4 points)

1. Define Active Share as introduced by Cremers and Petajisto (2009). (2 points)
2. What is a "closet indexer" and why should investors avoid them? (2 points)

Question 9.5 - MCQ: Fund Flows (2 points)

According to Frazzini and Lamont (2008), what does the "dumb money" effect refer to?

A) Funds receiving high inflows subsequently outperform
B) Retail investors reduce their wealth by reallocating across mutual funds
C) Professional investors consistently beat retail investors
D) Small funds perform worse than large funds

Question 9.6 - Smart Money vs. Dumb Money (6 points)

1. Zheng (Study of Mutual Fund Investors' Fund Selection Ability) finds evidence of "smart money" at the fund level. Explain what this means. (3 points)
2. Frazzini and Lamont find "dumb money" at the stock level. How can both findings be true simultaneously? (3 points)

Lecture 10: Empirical Corporate Finance & Event Studies

Question 10.1 - Event Study Methodology (8 points)

Describe the key steps in conducting an event study:

1. Event identification and event window definition (2 points)
2. Estimation window and normal returns (2 points)
3. Calculating abnormal returns (2 points)
4. Testing for statistical significance (2 points)

Question 10.2 - Abnormal Returns Calculation (5 points)

A stock has the following characteristics:

- Beta = 1.2
- Event day return = 3%
- Market return on event day = 1%
- Risk-free rate = 0.2%

Calculate the abnormal return using the market model. Show your work.

Question 10.3 - Cumulative Abnormal Returns (4 points)

1. Define Cumulative Abnormal Returns (CAR). (2 points)
2. If daily abnormal returns are [-0.5%, 2%, 1.5%, 0.8%, -0.3%] for days -2 to +2 around an event, calculate the CAR[-2, +2]. (2 points)

Question 10.4 - MCQ: Event Study Assumptions (2 points)

Which assumption is critical for event study methodology to correctly identify the effect of an event?

A) Markets are always efficient
B) The event is unanticipated by market participants
C) The stock has high trading volume
D) The firm is large and well-known

Lecture 11 & 12: AI and NLP in Finance & Exercise 9

Question 11.1 - AI vs. Machine Learning Definitions (4 points)

1. Define Artificial Intelligence (AI) and Machine Learning (ML). (2 points)
2. Explain the relationship between AI, ML, Deep Learning, and Neural Networks. (2 points)

Question 11.2 - Classification vs. Regression (4 points)

Provide two examples each of classification tasks and regression tasks in finance. Explain why each example belongs to the respective category.

Question 11.3 - Decision Trees (6 points)

1. Explain how a decision tree works for classification tasks. (3 points)
2. What is overfitting in the context of decision trees? (2 points)
3. How do random forests address the overfitting problem? (1 point)

Question 11.4 - Validation Methods (5 points)

Explain the difference between:

1. Resubstitution error (2 points)
2. Training/testing data split (3 points)

Why is resubstitution error problematic?

Question 11.5 - AI Biases (6 points)

Explain three pitfalls in using AI for financial research:

1. Survivorship bias (2 points)
2. Look-ahead bias (2 points)
3. Overfitting (2 points)

Question 11.6 - Transformer Models (5 points)

Describe the workflow of transformer-based language models (LLMs) in processing text:

1. Tokenization and embeddings (2 points)
2. Self-attention mechanism (2 points)
3. Text generation (1 point)

Question 11.7 - Sentiment Analysis (5 points)

1. Explain the traditional dictionary method for sentiment analysis (e.g., Loughran-McDonald). (2 points)
2. How can generative AI (e.g., ChatGPT) improve sentiment analysis compared to dictionary methods? (3 points)

Question 11.8 - MCQ: Supervised Learning (2 points)

Which of the following best describes supervised learning?

A) Learning patterns without labeled output data
B) Learning a function from input to output using labeled data
C) Learning only from positive examples
D) Learning without any training data

Question 11.9 - Research Application - Lopez-Lira and Tang (2023) (4 points)

Explain how ChatGPT was used to forecast stock price movements in Lopez-Lira and Tang (2023). What were the main findings regarding predictive power?

Question 11.10 - MCQ: Random Forests (2 points)

Why are random forests less prone to overfitting compared to individual decision trees?

A) They use fewer features
B) They combine multiple trees trained on bootstrap samples
C) They require less training data
D) They are simpler models

Question 11.11 - MSE Calculation (5 points)

Given predictions [2, 4, 6] and actual values [2.5, 3.5, 5.5]:

1. Calculate the Mean Squared Error (MSE). (4 points)
2. Interpret what MSE measures. (1 point)

Question 11.12 - Limitations of Generative AI (6 points)

Discuss three limitations of using generative AI in financial research:

1. Hallucinations (2 points)
2. Look-ahead bias in training data (2 points)
3. Low replicability (2 points)

Additional Mixed Questions Across Topics

Question 12.1 - Research Design (5 points)

You want to test whether firms with high R&D intensity earn higher returns. Describe:

1. The characteristic you would measure (1 point)
2. How you would form portfolios (2 points)
3. How you would test for statistical significance (2 points)

Question 12.2 - Data Quality (4 points)

Explain two reasons why survivorship bias in mutual fund databases leads to an overestimation of average fund performance.

Question 12.3 - Factor Model Comparison (6 points)

Compare the CAPM, Fama-French 3-factor, and Fama-French 5-factor models:

1. List the factors in each model (3 points)
2. What motivated the addition of factors beyond CAPM? (3 points)

Question 12.4 - MCQ: Market Anomalies (2 points)

Which of the following is an example of a momentum anomaly?

A) Stocks with high book-to-market ratios earn higher returns
B) Stocks that rose in the past 3-12 months continue to rise
C) Small stocks earn higher returns than large stocks
D) Highly profitable firms outperform less profitable firms

Question 12.5 - Industry Concentration (5 points)

Kacperczyk, Sialm, and Zheng (2005) find that mutual funds concentrated in fewer industries outperform diversified funds. Explain why industry concentration might indicate skill rather than luck.

Answer Key

Lecture 1 & Exercise 1: Introduction to Empirical Finance

Question 1.1: Capital markets solve the fundamental problem that those who have capital are not the ones who need it. Savers (individuals, pension funds) have excess capital but limited investment expertise, while firms and entrepreneurs need capital for investment opportunities but lack funds. Capital markets provide an efficient mechanism for transferring resources from savers to productive investments. Direct financing is more efficient for certain investments because it eliminates intermediary costs and allows for better risk-return matching between specific investors and projects.

Question 1.2:

- **Primary markets**: Where new securities are issued and sold for the first time (e.g., an IPO where a company sells new shares to raise capital).
- **Secondary markets**: Where existing securities are traded among investors (e.g., buying Apple stock on NASDAQ from another investor).
- Secondary markets are essential because they provide liquidity, making investors more willing to purchase securities in primary markets, knowing they can later sell them.

Question 1.3:

1. Sharpe ratio A = (8% - 2%) / 12% = 0.50; Sharpe ratio B = (6% - 2%) / 8% = 0.50
2. Both portfolios offer identical risk-adjusted performance (same Sharpe ratio)
3. Sharpe ratio measures excess return per unit of total risk (standard deviation), indicating how much additional return an investor receives for taking on extra volatility

Question 1.4: C) Weak-form efficiency (implies that past price information is already reflected in current prices, making technical

analysis ineffective)

Lecture 2 & Exercise 2: Financial Data Structures

Question 2.1:

1. **Share Type** = **"NS"**: Ensures only ordinary common shares are included. Without this, the sample would include certificates, ADRs, units, REITs, which have different risk-return characteristics and could distort cross-sectional analyses.
2. **US Incorporation Flag** = **"Y"**: Restricts sample to US-incorporated companies. Without this, foreign firms trading in US markets would be included, introducing different regulatory, tax, and accounting standards.
3. **Primary Exchange** = **"N", "A", "Q"**: Limits to major exchanges (NYSE, AMEX, NASDAQ) which cover >98% of US traded stocks. Omitting this would include OTC stocks with poor liquidity and less reliable pricing.

Question 2.2:

- **PERMNO**: Unique identifier for a security (stock). Each traded security gets one PERMNO.
- **PERMCO**: Unique identifier for a company/firm. One PERMCO can have multiple PERMNOs because a company may issue different share classes (e.g., Google has Class A and Class C shares, both under the same PERMCO but different PERMNOs).

Question 2.3:

1. Use END-of-January 2024 price and shares outstanding to calculate market cap for portfolio formation.
2. Using end-of-February prices would introduce look-ahead bias because at the end of January, you don't yet know what prices will be in February. Investment decisions must use only information available at the decision time.

Question 2.4: B) Returns are scale-free and comparable across stocks

Question 2.5:

1. Fiscal year-end dates don't align with calendar year-end because companies choose fiscal years based on business cycles (e.g., retailers often end in January after holiday season; Microsoft ends in June).
2. Multiple datadate entries may exist per fiscal year due to restatements or corrections. Using only the last available datadate ensures the most accurate and final reported numbers, avoiding duplicate observations.

Lecture 3 & Exercise 3: Mean-Variance Portfolio Theory

Question 3.1:

1. $E[R_p] = 0.6(10\%) + 0.4(6\%) = 6\% + 2.4\% = 8.4\%$
2. $\sigma_p^2 = (0.6)^2(0.20)^2 + (0.4)^2(0.10)^2 + 2(0.6)(0.4)(0.3)(0.20)(0.10)$
   $= 0.36(0.04) + 0.16(0.01) + 2(0.24)(0.006)$
   $= 0.0144 + 0.0016 + 0.00288 = 0.01888$
3. $\sigma_p = \sqrt{0.01888} = 13.74\%$

Question 3.2:

1. The efficient frontier is the upper boundary of the feasible set of portfolios in mean-variance space, showing portfolios with maximum expected return for each level of risk, or minimum risk for each level of expected return. Graphically, it's an upward-sloping convex curve in (σ, E[R]) space.
2. Rational investors never choose portfolios below the efficient frontier because for the same level of risk, they could achieve higher expected return by selecting a portfolio ON the frontier (dominance principle).

Question 3.3:
$w_1 = (0.01 - 0.01) / (0.04 + 0.01 - 2(0.01))$
$= 0 / (0.05 - 0.02)$
$= 0 / 0.03 = 0$

Weight in Asset 1 = 0%, Weight in Asset 2 = 100%

Question 3.4: B) The optimal trade-off between risk and return using the risk-free asset and the tangency portfolio

Question 3.5: Short selling involves borrowing a security and selling it, hoping to buy it back later at a lower price. A negative portfolio weight means you're short that asset (selling borrowed shares). This violates the "no short sales" constraint ($w_i \geq 0$) in basic mean-variance optimization, though it's allowed in extended models.

Lecture 4 & Exercise 4: CAPM and Factor Models

Question 4.1:

1. $\beta = Cov(R_a, R_m) / Var(R_m) = 0.024 / 0.04 = 0.6$
2. $E[R_a] = R_f + \beta(E[R_m] - R_f) = 3\% + 0.6(9\% - 3\%) = 3\% + 3.6\% = 6.6\%$
3. α = Realized return - Expected return = 8% - 6.6% = 1.4%

Question 4.2: Four key CAPM assumptions:

1. **Investors are rational and risk-averse**: Realistic for institutional investors, but behavioral biases affect retail investors.
2. **No transaction costs or taxes**: Unrealistic; trading costs and taxes affect all investors.
3. **All investors have identical time horizons**: Unrealistic; investors have different investment horizons.
4. **All investors have homogeneous expectations**: Unrealistic; investors have different views about expected returns and risks.

Question 4.3:

1. **Systematic risk**: Risk that affects all securities and cannot be eliminated through diversification (e.g., market risk, economic cycles). **Idiosyncratic risk**: Firm-specific risk that can be diversified away (e.g., CEO departure, product recall).
2. Only systematic risk is rewarded with a risk premium because idiosyncratic risk can be eliminated through diversification, so rational investors should not demand compensation for bearing it.

Question 4.4:

1. **SMB (Small Minus Big)**: The return difference between small-cap and large-cap stocks, capturing the size effect. **HML (High Minus Low)**: The return difference between high book-to-market (value) and low book-to-market (growth) stocks, capturing the value effect.
2. These factors were added because empirical evidence showed that CAPM's single market factor couldn't explain cross-sectional variation in returns—size and value effects persisted even after controlling for market beta.

Question 4.5: C) Systematic risk that cannot be diversified away

Lecture 5 & Exercise 5: Behavioral Finance

Question 5.1: Three behavioral biases:

1. **Overconfidence**: Investors overestimate their knowledge, abilities, and the precision of their information. Example: An investor trades excessively because they believe they can predict market movements better than others, leading to high transaction costs and poor performance.
2. **Herding**: Investors follow the decisions of others rather than their own analysis. Example: During the dot-com bubble, investors bought internet stocks simply because others were buying, ignoring fundamental valuations.
3. **Home bias**: Investors disproportionately invest in domestic assets despite benefits of international diversification. Example: US investors hold 70%+ of portfolios in US stocks despite the US representing only ~50% of global market cap.

Question 5.2:

1. Overconfidence bias occurs when investors overestimate their ability to select winning investments, time the market, or process information better than others.
2. Overconfident investors trade more frequently, believing each trade will be profitable. This leads to excessive transaction costs that erode returns. They also underestimate risks, creating poorly diversified portfolios that underperform.

Question 5.3:

1. Expected utility theory assumes people evaluate outcomes based on final wealth levels with consistent risk preferences. Prospect theory shows people evaluate outcomes as gains/losses relative to a reference point and exhibit loss aversion and probability weighting.
2. Loss aversion means losses hurt about twice as much as equivalent gains feel good. This causes investors to hold losing positions too long (hoping to avoid realizing losses) and sell winners too early (to lock in gains), leading to the disposition effect.

Question 5.4: B) Sell winning stocks too quickly and hold losing stocks too long

Question 5.5: Two limits to arbitrage:

1. **Fundamental risk**: Even if an asset is mispriced, its price may diverge further before converging to fundamental value, causing losses for arbitrageurs with finite horizons or leverage constraints.
2. **Noise trader risk**: Irrational investors may push prices further from fundamentals, making arbitrage risky and requiring long holding periods that arbitrageurs may not be able to sustain.

Lecture 6 & Exercise 6: Performance Evaluation

Question 6.1:

1. Jensen's $\alpha = R_p - R_f - \beta(R_m - R_f) = 1.2\% - 0.3\% - 1.3(0.6\%) = 0.9\% - 0.78\% = 0.12\%$ per month
2. The fund outperformed by 0.12% per month (1.44% annually) after adjusting for systematic risk.
3. No, with $t = 1.5 < 1.96$, the alpha is not statistically significant at the 5% level.

Question 6.2:

1. Information Ratio = $\alpha / \sigma(\varepsilon)$, where α is the portfolio's alpha and $\sigma(\varepsilon)$ is the tracking error (standard deviation of residuals from the factor model).

2. Unlike Sharpe ratio (which uses total risk), Information Ratio measures risk-adjusted active return relative to a benchmark. It's more appropriate for evaluating active managers who track a benchmark closely.

Question 6.3: Treynor ratio measures excess return per unit of systematic risk (beta), while Sharpe ratio uses total risk (standard deviation). Treynor is more appropriate when evaluating a portfolio that's part of a larger, diversified portfolio, where only systematic risk matters.

Question 6.4: B) Jensen's alpha

Question 6.5:

1. Tracking error is the standard deviation of the difference between portfolio returns and benchmark returns, measuring active risk.
2. Information Ratio = α / Tracking Error → 0.5 = 2% / TE → TE = 2% / 0.5 = 4% per year

Lecture 7 & Exercise 7: Trading Strategies and Anomalies I

Question 7.1: Four-step framework:

1. **Step 1 - Rank stocks**: Sort all stocks by a characteristic (e.g., size, value, momentum) at time t.
2. **Step 2 - Form portfolios**: Group ranked stocks into portfolios (e.g., quintiles, deciles) based on breakpoints.
3. **Step 3 - Compute returns**: Calculate value-weighted or equal-weighted returns for each portfolio over the holding period (t to t+1).
4. **Step 4 - Test for abnormal performance**: Run regressions of long-short portfolio returns on risk factors (CAPM, FF3, etc.) and test if alpha is significantly different from zero.

Question 7.2:

1. A zero-investment long-short portfolio involves going long $X in high-characteristic stocks and short $X in low-characteristic

stocks, requiring no net capital (net investment = $0).

2. It's market-neutral because if the market moves up or down by X%, both long and short positions are affected equally, canceling out market exposure. The return captures only the characteristic effect.
3. Long-short return = 15% - 5% = 10%

Question 7.3:

- **Independent sorts**: Sort stocks by characteristic 1, then independently sort by characteristic 2. Use when testing the unconditional effect of each variable (e.g., size and value effects separately). Example: Fama-French size and value portfolios.
- **Dependent sorts**: First sort by characteristic 1, then within each group, sort by characteristic 2. Use when testing if one effect varies by another (e.g., does value work better in small or large stocks?). Example: Testing if momentum is stronger in small caps.

Question 7.4:

1. **Step 1 - Cross-sectional regressions**: Each month t, regress stock returns on lagged characteristics ($R_{it} = \alpha_{0t} + \beta_{1t}X_{1,i,t-1} + ... + \varepsilon_{it}$) across all stocks. Obtain coefficient estimates β_{1t}, β_{2t}, etc. for that month.
2. **Step 2 - Time-series averaging**: Compute the average coefficient across all months: $\bar{\beta}_1 = (1/T)\Sigma\beta_{1t}$. Calculate standard errors and t-statistics to test if $\bar{\beta} \neq 0$.
3. This two-step procedure is necessary to avoid time-series correlation in residuals, which would invalidate standard errors in a pooled panel regression.

Question 7.5: B) NASDAQ stocks are typically smaller, and using all-stock breakpoints would create size-biased portfolios

Lecture 8: Trading Strategies and Anomalies II

Question 8.1:

1. The "factor zoo" refers to the proliferation of hundreds of documented factors/anomalies that supposedly predict stock returns, growing from 5-10 in the 1970s to over 300 by the 2010s.
2. Two reasons for growth: (a) **Data availability and computational power**: Easier to test many strategies with modern computers and databases. (b) **Publication bias and academic incentives**: Journals favor novel, statistically significant findings, encouraging researchers to search for new factors.

Question 8.2:

1. P-hacking (data mining) is manipulating analysis—through variable selection, sample periods, or methodology choices—until desired statistical significance (p-value < 0.05) is achieved.
2. Examples: (a) Testing many portfolio formation rules (quintiles vs deciles, NYSE vs all-stock breakpoints, value-weighted vs equal-weighted) and reporting only the one yielding significant results. (b) Trying multiple time periods and excluding periods where the strategy doesn't work, then claiming the strategy "works" based on the cherry-picked sample.

Question 8.3: With hundreds of factors being tested, the probability of finding spurious significance by chance increases dramatically (multiple testing problem). The traditional t > 1.96 threshold (5% significance) was designed for testing a single hypothesis. Harvey et al. show that with extensive data mining, a much higher hurdle (t > 3.0) is needed to maintain proper Type I error control across multiple tests.

Question 8.4:

1. Two explanations: (a) **Arbitrage/informed trading**: Sophisticated investors trade on published anomalies,

eliminating mispricing. (b) **Data mining/overfitting**: Original findings were inflated by p-hacking; strategies never truly worked out-of-sample.
2. The arbitrage explanation is more consistent with market efficiency—markets learn and correct mispricing once it becomes public.

Question 8.5: D) 82%

Lecture 9 & Exercise 8: Mutual Funds and Active Management

Question 9.1:

1. **Trading/pricing**: Mutual funds priced once daily at NAV; investors buy/redeem directly with fund. ETFs trade continuously on exchanges like stocks at market prices; investors trade with other investors.
2. **Fees**: Mutual funds typically have higher expense ratios (0.5-1.5%), often with load fees. ETFs have lower expense ratios (0.05-0.50%) and no load fees.
3. **Tax efficiency**: Mutual funds distribute capital gains when portfolio managers sell securities, creating taxable events. ETFs use in-kind redemption, minimizing capital gains distributions, making them more tax-efficient.

Question 9.2:

1. When a skilled manager is identified (through past performance), rational investors allocate capital to that fund, increasing assets under management (AUM).
2. As AUM grows, the manager faces decreasing returns to scale—good ideas become harder to implement at larger size, and alpha declines toward zero.
3. Investors still allocate capital because skilled managers earn fees on AUM even when alpha = 0. The manager captures the value of skill through fees, not through alpha available to investors.

Question 9.3:

1. This implies that in aggregate, managers have zero skill before fees. Any skill that exists is exactly offset by negative skill or luck, and after deducting fees, investors are worse off than passive indexing.
2. To distinguish luck from skill: (a) Test for persistence—skilled managers should consistently outperform over multiple periods. (b) Use bootstrap simulations to see if observed performance distribution differs from what random chance would produce. (c) Examine performance in extreme tails (very good and very bad funds) where skill is more likely to show.

Question 9.4:

1. Active Share measures the percentage of portfolio holdings that differ from the benchmark index. Active Share = 50% means half the portfolio differs from the index.
2. A "closet indexer" is a fund with low Active Share (< 20%) that closely tracks the index while charging active management fees. Investors should avoid them because they pay high fees for essentially passive index-like performance.

Question 9.5: B) Retail investors reduce their wealth by reallocating across mutual funds

Question 9.6:

1. "Smart money" at fund level means investors successfully identify skilled funds—funds receiving high inflows subsequently outperform, suggesting investors correctly infer manager skill from past performance.
2. Both can be true because: At the fund level, informed investors pick good managers early (first-mover advantage). At the stock level, aggregate retail flows chase glamour stocks at market peaks, exhibiting behavioral biases (extrapolation, representativeness), buying high and selling low. Fund-level allocation is more sophisticated than stock-level timing.

Lecture 10: Empirical Corporate Finance & Event Studies

Question 10.1:

1. **Event identification and window**: Identify precise event date (announcement/occurrence); define event window (e.g., [-1, +1] days around event) when impact is measured.
2. **Estimation window and normal returns**: Select clean estimation window before event (e.g., [-250, -11] days); estimate normal return model (e.g., market model: $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$).
3. **Abnormal returns**: Calculate abnormal return as AR = Actual return - Expected normal return for each day in event window.
4. **Statistical significance**: Compute cumulative abnormal returns (CAR) and test using t-statistics whether CAR is significantly different from zero, indicating market reaction to event.

Question 10.2:
Expected return = $R_f + \beta(R_m - R_f)$ = 0.2% + 1.2(1% - 0.2%) = 0.2% + 0.96% = 1.16%
Abnormal return = Actual return - Expected return = 3% - 1.16% = 1.84%

Question 10.3:

1. CAR (Cumulative Abnormal Return) is the sum of abnormal returns over a specified event window, measuring the total market reaction to an event.
2. CAR[-2, +2] = -0.5% + 2% + 1.5% + 0.8% + (-0.3%) = 3.5%

Question 10.4: B) The event is unanticipated by market participants

Lecture 11 & 12: AI and NLP in Finance & Exercise 9

Question 11.1:

1. **AI**: Technology enabling computers to simulate human learning, problem-solving, and decision-making. **ML**: A branch of AI using

data and algorithms to enable systems to learn and improve accuracy without explicit programming.
2. AI is the broadest concept. ML is a subset of AI. Deep Learning is a subset of ML using multi-layered neural networks. Neural Networks are the computational models used in Deep Learning.

Question 11.2:
Classification examples: (1) Predicting bond default (yes/no), (2) Detecting positive/negative sentiment in financial news. These predict discrete categories.
Regression examples: (1) Forecasting exact stock return percentage, (2) Estimating future inflation rate. These predict continuous numerical values.

Question 11.3:

1. A decision tree learns rules from training data by recursively splitting features to minimize prediction error. For classification, it splits based on features that best separate classes. Each branch represents a decision rule; leaf nodes represent class predictions.
2. Overfitting occurs when a tree becomes too complex (too many splits/deep), memorizing training data including noise, leading to poor generalization on new data.
3. Random forests build many trees on bootstrap samples of data, with each split considering only a random subset of features. Final prediction is by majority vote, reducing variance and overfitting.

Question 11.4:

1. **Resubstitution error**: Using the same data for training and testing; comparing predictions to actual values in training set. Problematic because it overestimates model performance (model has already seen the data).
2. **Training/testing split**: Randomly split data (e.g., 80/20 or 70/30) into training set (fit model) and testing set (evaluate performance). Provides honest out-of-sample performance estimate because testing data was never seen during training.

Question 11.5:

1. **Survivorship bias**: Using only firms/funds that survived the observation period, excluding failures/delistings. This inflates returns because failed entities (which typically had poor performance) are omitted.
2. **Look-ahead bias**: Using information not available at the decision time (e.g., using future earnings to predict current returns). This artificially inflates model performance.
3. **Overfitting**: Model becomes too complex, fitting noise in training data rather than true patterns, leading to poor out-of-sample performance despite excellent in-sample fit.

Question 11.6:

1. **Tokenization and embeddings**: Text is split into tokens (words/subwords); each token is converted to a numerical vector capturing semantic meaning.
2. **Self-attention mechanism**: Model calculates relationships between all tokens, determining which words are most relevant to each other, capturing context and meaning.
3. **Text generation**: Model outputs probability distributions over possible next tokens; highest probability token is selected (or sampled) iteratively to generate text.

Question 11.7:

1. Dictionary method counts positive and negative words (e.g., "profit," "growth" vs. "loss," "decline") in text. More positive words → positive sentiment; more negative words → negative sentiment.
2. Generative AI improves by: (a) Understanding context—"not good" is negative despite containing "good". (b) Capturing nuance and sarcasm that word-counting misses. (c) Processing longer documents without losing context through token limits.

Question 11.8: B) Learning a function from input to output using labeled data

Question 11.9:
ChatGPT was prompted to analyze news headlines and predict whether stock prices would go up, down, or stay neutral based on sentiment. Main findings: ChatGPT scores partially predicted out-of-

sample daily returns, especially for smaller stocks and following negative news. Performance improved with model sophistication (GPT-4 > simpler models).

Question 11.10: B) They combine multiple trees trained on bootstrap samples

Question 11.11:

1. Errors: [2-2.5, 4-3.5, 6-5.5] = [-0.5, 0.5, 0.5]
   Squared errors: [0.25, 0.25, 0.25]
   MSE = (0.25 + 0.25 + 0.25) / 3 = 0.75 / 3 = 0.25
2. MSE measures average squared prediction error, penalizing larger errors more heavily. Lower MSE indicates better predictive accuracy.

Question 11.12:

1. **Hallucinations**: AI generates plausible but incorrect information, presenting false data as factual. In finance, this could mean fabricated financial metrics or nonexistent research findings.
2. **Look-ahead bias in training**: Models trained on data including future information may appear to predict well in-sample but fail out-of-sample. Difficult to detect without careful data construction.
3. **Low replicability**: Results vary across model runs, configurations, and datasets due to stochastic training processes. Exact outcomes difficult to reproduce, violating scientific replication standards.

Additional Mixed Questions

Question 12.1:

1. Measure R&D intensity as R&D expense / Total assets or R&D / Sales using Compustat annual data.
2. Each June, sort stocks into quintiles based on R&D intensity from previous fiscal year; form value-weighted or equal-weighted portfolios; hold for 12 months (July-June).

3. Regress long-short (high-low R&D) portfolio returns on FF3 or FF5 factors; test if alpha is statistically significant using t-statistic ($|t| > 1.96$ for 5% level, or $|t| > 3.0$ for multiple testing adjustment).

Question 12.2:

1. Failed funds (poor performers) are removed from databases when they close/merge, leaving only survivors (better performers). This overstates average historical performance.
2. Survivor funds often had period of strong performance before being included in database (incubation bias), while failed funds' poor early performance is excluded, creating upward bias in measured returns.

Question 12.3:

1. **CAPM**: Market factor (R_m - Rբ). **FF3**: Market, SMB (size), HML (value). **FF5**: Market, SMB, HML, RMW (profitability), CMA (investment).
2. CAPM couldn't explain cross-sectional return patterns—small stocks and value stocks earned higher returns than beta predicted. FF3 added size and value factors. FF5 added profitability and investment factors after research showed these characteristics predict returns beyond FF3.

Question 12.4: B) Stocks that rose in the past 3-12 months continue to rise

Question 12.5:
Industry concentration indicates skill because: (1) Managers with deeper knowledge of specific industries have informational advantages—better monitoring, superior valuation, industry-specific expertise. (2) Concentration reflects conviction; managers invest more in their best ideas where they have edge, rather than diversifying into areas where they have no advantage. (3) The 1-2% annual outperformance persists after controlling for risk, suggesting genuine stock-selection ability rather than random luck.